

will be of the form

$$\mathcal{L} = \left[\Phi^\dagger e^{-V} \Phi \right]_D + 2 \operatorname{Re} \left[f(\Phi) \right]_{\mathcal{F}} + \frac{1}{2g^2} \operatorname{Re} \left[\sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}, \quad (27.6.1)$$

where the superpotential $f(\Phi)$ is a gauge-invariant cubic polynomial in the left-chiral superfields. (We are ignoring a possible θ -term, which has no effect in perturbation theory.)

Suppose we impose an ultraviolet cut-off λ on the momenta circulating in loop graphs. As discussed in Section 12.4, we can find a *local* ‘Wilsonian’ effective Lagrangian density $\mathcal{L}_\lambda$ that, with this cut-off, gives precisely the same results as the original Lagrangian density for S -matrix elements of processes at momenta below λ . The effective Lagrangian density has masses and coupling parameters that now depend on λ , and usually there will be an infinite number of coupling terms in the effective Lagrangian density, all possible terms allowed by the symmetries of the theory. But things are much simpler in supersymmetric theories. The non-renormalization theorems tell us that, as long as the cut-off preserves supersymmetry and gauge invariance, to all orders in perturbation theory the effective Lagrangian will have the structure

$$\begin{aligned} \mathcal{L}_\lambda = & \left[\mathcal{A}_\lambda(\Phi, \Phi^\dagger, V, \mathcal{D} \cdots) \right]_D + 2 \operatorname{Re} \left[f(\Phi) \right]_{\mathcal{F}} \\ & + \frac{1}{2g_\lambda^2} \operatorname{Re} \left[\sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}, \end{aligned} \quad (27.6.2)$$

where $\mathcal{A}_\lambda$ is a general Lorentz- and gauge-invariant function; ‘ $\mathcal{D} \cdots$ ’ denotes terms involving superderivatives or spacetime derivatives of the preceding arguments; and g_λ is the *one-loop* effective gauge coupling, given by the same formula as the one-loop renormalized gauge coupling constant

$$g_\lambda^{-2} = \text{constant} - 2b \ln \lambda, \quad (27.6.3)$$

where b is the coefficient of g^3 in the Gell-Mann–Low function $\beta(g)$, discussed in Chapter 18. This is for a simple gauge group, with a single gauge coupling, but the extension to a direct product of simple and $U(1)$ gauge groups is trivial. Note in particular that the effective superpotential is not only finite in the limit $\lambda \rightarrow \infty$, but at least in perturbation theory it contains no terms beyond those in the original superpotential, and there is no change in the coefficients of the terms it does contain.

To prove this theorem, we shall interpret this theory as a special case of one with two additional external gauge-invariant left-chiral superfields